An Upper Bound for the Circular Chromatic Number of Mycielski Graphs*

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Abstract

In a search for triangle-free graphs with arbitrarily large chromatic numbers, Mycielski ([15]) developed a graph transformation that transforms a graph G into a new graph M(G), we now call the Mycielskian of G. For $t \geq 2$, $M^t(G) = M(M^{t-1}(G))$. *Lam et al ([12]) generalized the definition to get the m-Mycielskian graph* ($\mu_m(G)$). *The problem of determining the circular chromatic numbers of these graphs has been investigated in many papers. In this paper, we shall study the range of* $\chi_c(M^t(G))$, *especially for* $G=K_n$ *or* K_d^k. *These investigations lead to an upper bound for the Mycielskians: if* $\chi_c(G) \leq (\chi(G)-1)+1/3$, *then* $\chi_c(M^{2t}(G)) \leq (\chi(M^{2t}(G))-1)+1/3$ *for every positive integer t.*

1. Introduction

All graphs considered are finite and simple. The circular chromatic number $\chi_c(G)$ of a graph G, introduced by Vince [18] in 1988 (under the name "star chromatic number"), is a natural generalization of the chromatic number of a graph (cf. [20]). For a pair of positive integers k and d with $k \geq 2d$, a (k, d)-*coloring* of a graph G is a mapping c of $V(G)$ to the set $\{0, 1, \ldots, k-1\}$ such that for any adjacent vertices x and y of G, $d \leq |c(x) - c(y)| \leq k - d$. The *circular chromatic number* $\chi_c(G)$ of a graph G is the infimum of the ratios k/d for which there exists a (k,d)-coloring of G.

It is easy to see that a $(k,1)$-coloring of a graph G is just the ordinary k-coloring of G. Therefore, $\chi_c(G) \leq \chi(G)$. On the other hand, it was proved in [18] that $\chi(G)-1 < \chi_c(G)$. Thus $\chi(G) = \lceil \chi_c(G) \rceil$. It is known [18] that there are graphs G with $\chi_c(G) = \chi(G)$ as well as graphs with $\chi_c(G)$ arbitrarily close to $\chi_c(G)-1$.

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For a graph G with vertex set $V(G)=V$ and edge set $E(G)=E$, the *Mycielskian* of G, denoted by $M(G)$, is the graph with vertex set $V \cup V' \cup \{u\}$, where $V'=\{x': x \in V\}$ and edge set $E \cup \{x'y: xy \in E\} \cup \{x'u: x' \in V'\}$. The vertex x' is called the *twin* of x (and x is also the twin of x'). The vertex u is called the *root* of $M(G)$. For $t \geq 2$, $M^t(G) = M(M^{t-1}(G))$ is called the *t-th Mycielskian* of graph G.

It is well known that for any nonempty graph G, $\chi(M(G)) = \chi(G) + 1$ and hence $\chi(M^t(G)) = \chi(G) + t$. However, there is no simple formula for $\chi_c(M(G))$ in terms of $\chi_c(G)$. The problem of determining the circular chromatic number of the iterated Mycielskian of graphs has been investigated in [3, 4, 10, 11, 12, 14]. But those results are still just a fraction of the properties concerning the circular chromatic number of Mycielski's graphs

Let K_n denote the complete graph on n vertices. For two positive integers k and d such that $k \geq 2d$, K_d^k (denoted as G_k^d previously) is the graph with vertex set $\{0, 1, 2, \ldots, k-1\}$ in which ij is an edge if and only if $d \leq |i-j| \leq k-d$. It is known that the circular chromatic number $\chi_c(K_d^k) = k/d$, and any graph is (k, d)-colorable if and only if there exists a homomorphism from G to K_d^k. Therefore, in the study of circular chromatic numbers, the graphs K_d^k play the same role that complete graphs K_n do in the study of chromatic numbers. The graph K_d^k is naturally called a *circular complete graph*.

Recently, the study of the circular chromatic number of a Mycielski's graph is focus on determining whether $\chi_c(M^t(K_n)) = \chi(M^t(K_n))$, and $\chi_c(M^t(K_d^k)) = \chi(M^t(K_d^k))$. In this paper, we shall refine some tools used by others in the study of the relationship between the circular chromatic number and the chromatic number of a graph.

2. Upper bounds of *d* when $\chi_c(M^t(G)) = k/d$

Let G be a graph with circular chromatic number k/d, $\gcd(k,d)=1$. In this section, we shall discuss the possible values of d, especially when G is the Mycielskian of some graph.

Lemma 2.1 (Fan [4]) If $\chi_c(G) = k/d$, gcd(k,d) =1, then $\deg_G(v) \le |V(G)|-2d+1$ for any $v \in V(G)$.

Let $\Delta(G)$ be the maximum degree of G. If $\chi_c(G) = k/d$ for a Mycielskian graph G with gcd(k, d) =1, then the range of d can be limited more than a general graph since $\Delta(G)$ must be large.

Theorem 2.2 Let G be a Mycielskian graph. If $\chi_c(G) = k/d$, gcd(k, d) =1, then $d \le (|V(G)| + 3)/4$.

Proof. By Lemma 2.1, $\Delta(G) \le |V(G)| - 2d + 1$. We get $d \le (|V(G)| + 1 - \Delta(G))/2$, Since the root vertex of G has degree $(|V(G)| - 1)/2$. It follows that $\Delta(G) \ge (|V(G)| - 1)/2$ and hence $d \le (|V(G)| + 3)/4$. ■

Theorem 2.3 Let $M^t(G)$ be the t-th Mycielskian of a graph G. If $\chi_c(M^t(G)) = k/d$ with gcd(k,d) = 1, then $d \le (2^{t-1})(|V(G)|+ 1-\Delta(G))$.

Proof. By Lemma 2.1, $d \le (|V(M^t(G)| + 1 - \deg_{M^t(G)}(v)) / 2$ for each $v \in V(M^t(G))$. For a vertex w of G, if $\deg_G(w) = a$, then $\deg_{M(G)}(w) = 2a$, and hence $\deg_{M^t(G)}(w) = 2^t a$. Since $|V(M^t(G))| = 2^t|V(G)|+2^{t-1} +2^{t-2} + \dots +1 = 2^t|V(G)|+2^t-1$, we get $d \le (2^t|V(G)| + 2^t - 2^t\Delta(G)) / 2 = (2^{t-1})(|V(G)| + 1 - \Delta(G))$. ■

Corollary 2.4 Let G be a graph with a universal vertex. If $\chi_c(M^t(G)) = k/d$, gcd(k,d) =1, then $d \le 2^t$. Particularly, $\chi_c(G) = \chi(G)$, $\chi_c(M(G)) = \chi(M(G))$ or $\chi(M(G)) - 1/2$.

Proof. Since the graph has $\Delta(G) = |V(G)| - 1$. It follows that $d \le (2^{t-1})(|V(G)| + 1 - \Delta(G)) = 2^t$ by Theorem 2.3. Moreover, if $t = 0$ then $d = 1$ and hence $\chi_c(G) = \chi(G)$. If $t = 1$ then $d \le 2$ and hence $\chi_c(M(G)) = \chi(M(G))$ or $\chi(M(G)) - 1/2$. ■

Consider the graphs $M^t(K_n)$, the upper bound of d can be improved by the following lemma.

Lemma 2.5 (Bondy [2]) Let G be a graph with circular chromatic number $\chi_c(G) = k/d$, gcd(k,d) =1, then $k \le |V(G)|$.

Theorem 2.6 If $\chi_c(M^t(K_n)) = k/d$, gcd(k,d) =1, then $d \le (2^t n+2^t-1)/(n+t-1)$.

Proof. Since $|V(K_n)| = n$, $|V(M^t(K_n))| = 2^t n+ 2^t-1$. If $\chi_c(M^t(K_n)) = k/d$ with gcd(k,d) =1, then $k \le 2^t n+ 2^t-1$ by Lemma 2.5, and $k/d > n+t-1$ since $\chi(M^t(K_n)) = n+t$. Therefore $d < k/(n+t-1) \le (2^t n+2^t-1)/(n+t-1)$. ■

Note that $(2^t n+2^t-1)/(n+t-1) < 2^t$ when $t \ge 2$. The following result for $M^2(K_n)$ then derived from above theorem.

Corollary 2.7 If $\chi_c(M^2(K_n)) = k/d$ with gcd(k, d) =1, then $d \le 3$.

It is known ([3]) that $\chi_c(M^2(K_3)) \le 9/2$ and $\chi_c(M^3(K_4)) \le 13/2$. This corollary tell us $\chi_c(M^2(K_3))$ is either 9/2 or 13/3. Since no (13,3)-coloring of $M^2(K_3)$ can be found by computers, we believe that $\chi_c(M^2(K_3)) = 9/2$. Similarly, $\chi_c(M^3(K_4))$ may be one of 6+1/2, 6+1/3, 6+1/4, 6+1/5, 6+2/5, and 6+1/6. Since no (32, 5)-coloring of $M^3(K_4)$ can be found by computers, we believe that $\chi_c(M^3(K_4)) = 13/2$.

Hajiabolhassan and Zhu ([10]) offered another upper bound of d. Suppose $G = (V, E)$ is a graph. For a vertex x of G, we denote by $\overline{N}_G(x)$ the set of non-neighbors of x, i.e., $\overline{N}_G(x) = \{ y \in V(G): xy \notin E(G), y \ne x\}$. For any set $X \subseteq V(G)$, let $\overline{N}_G(X) = \bigcup_{x\in X} \overline{N}_G(x)$, the notation X is also used to denote the subgraph of G induced by X. A set $Y \subseteq \overline{N}_G(X)$ is called a *pointwise-dominating set* of $\overline{N}_G(X)$ if for each $x \in X$, $\overline{N}_G(x) - Y$ is an independent set. A parameter $\beta_G(X)$ is defined as follows:
$\beta_G(X) = \min\{|Y| : Y$ is a pointwise-dominating set of $\overline{N}_G(X)\}$.

Lemma 2.8 (Hajiabolhassan and Zhu [10]) Suppose G is a graph, $t \ge 1$, and $\chi_c(M^t(G)) = k/d$ (where gcd(k,d) =1). If X is a clique of G, then $(|X| - 3)(d - 1) \le 2^t\beta_G(X) + 2^t - 2$.

Therefore $d \le (2^t\beta_G(X) + 2^t - 2)/(|X| - 3) + 1$. Since $\beta_G(X) = 0$ for a complete graph K_n, $d \le (2^t - 2)/(n - 3) +1$. So Hajiabolhassan and Zhu proved that $\chi_c(M^t(K_n)) = \chi(M^t(K_n))$ if $n \ge 2^t + 2$. Let us consider the graph $M^t(K^k_d)$ where K^k_d is not a complete graph.

Lemma 2.9 Let K^k_d be a circular complete graph with $sd < k < (s+1)d$, where $d \ge 2$, $s \ge 2$, and gcd(k, d) = 1. There is a maximum clique X of K^k_d with $|X| = s$ and $\beta_{K^k_d}(X) \le (d-1)\lceil s/2\rceil$.

Proof. Let X be the set $\{0, d, 2d, \dots, (s-1)d\}$. It is clear that X forms a clique of size s. Since $\chi_c(K^k_d) = k/d < s+1$, K^k_d can not have a larger clique. Therefore X is a maximum clique. Since for each vertex $x \in X$, $\overline{N}_G(x) = \{x-d+1, x-d+2, \dots, x, x+1, \dots, x+d-1\}$ (the values should take module k). It is clear that $\overline{N}_G(X) = V(K^k_d) = \{0, 1, \dots, k-1\}$. Moreover, since $\overline{N}_G(x)$ has $2d - 1$ consecutive integers, a pointwise-dominating set Y should satisfy that $|\overline{N}_G(x) - Y| \le d$ for each $x \in X$. It follows that $|\overline{N}_G(x) \cap Y| \ge d - 1$. Let $Y = \{y \in V(K^k_d): y = 2id + j$, where $0 \le i \le \lceil s/2\rceil - 1$ and $1 \le j \le d - 1\}$. It is easy to see that Y is a pointwise-dominating set of $\overline{N}_G(X)$. Therefore $\beta_{K^k_d}(X) \le (d-1)\lceil s/2\rceil$. ■

By Lemma 2.8 and Lemma 2.9, we have the following result.

Theorem 2.10 Let K_d^k be a circular complete graph with $sd < k < (s+1)d$, where $d \geq 2$, $s \geq 4$, and $\gcd(k, d) = 1$. Let $t \geq 1$. If $\chi_c(M^t(K_d^k)) = k'/d'$ with $\gcd(k', d') = 1$, then $d' \leq (2^t(d-1)\lceil s/2 \rceil + 2^t - 2)/(s-3) + 1$.

Corollary 2.11 Let K_d^k be a circular complete graph with $sd < k < (s+1)d$, where $d \geq 2$, $s \geq 4$, and $\gcd(k, d) = 1$. If $\chi_c(M(K_d^k)) = k'/d'$ with $\gcd(k', d') = 1$, then $d' \leq 2(d-1)\lceil s/2 \rceil/(s-3) + 1$.

If $d = 2$ with large s, the upper bound of d can be reduced as follows:

Corollary 2.12 If $\chi_c(M^t(K_2^{2s+1})) = k'/d'$ with $\gcd(k', d') = 1$, and $s > 3 \cdot 2^t + 1$, then $d' \leq 2^{t-1} + 1$.

Proof. Since $d = 2$, $d' \leq (2^t\lceil s/2 \rceil + 2^t - 2)/(s-3) + 1 \leq (2^{t-1}(s+1) + 2^t - 2)/(s-3) + 1 < 2^{t-1} + 2$. ■

If $t = 1$, Corollary 2.12 implies that $d' \leq 2$ when $s \geq 8$. Since it is known that $\chi_c(M(K_d^k)) = \chi(M(K_d^k))$ (showed by Huang and Chang [11]), the result may not be helpful. But if $t = 2$, Corollary 2.12 implies that $d' \leq 3$ when $s \geq 14$. Since it is proved that $\chi_c(M^2(K_2^{2s+1})) \leq s + 2 + 1/2$ in [3] (see Theorem 3.1 in the next section). The following corollary then derived:

Corollary 2.13 If $s \geq 14$, $\chi_c(M^2(K_2^{2s+1})) = s+2+1/2$ or $s+2+1/3$.

3. The graphs $M^t(K_d^k)$

In this section, we shall discuss the circular chromatic numbers of the graph $M^t(K_d^k)$. There are two main results for these graphs: Huang and Chang [11] showed that $\chi_c(M(K_d^k)) = \chi(M(K_d^k))$; Chang et al [3] proved the following result:

Theorem 3.1. (Chang [3]). If G is a graph for which $\chi_c(G) \leq \chi(G) - 1/d$, $d = 2$ or 3, then $\chi_c(M^2(G)) \leq \chi(M^2(G)) - 1/d$. Moreover, $\chi_c(M^{2t}(G)) \leq \chi(M^{2t}(G)) - 1/d$.

This theorem showed that if $\chi_c(G) \leq \chi(G) - 1/3$, or we can say that if $\chi_c(G)$ and $\chi(G)$ are not too close, then $\chi_c(M^{2t}(G))$ and $\chi(M^{2t}(G))$ cannot be equal. On the other hand, we can show that if $\chi_c(G)$ and $\chi(G) - 1$ are close enough, then $\chi_c(M^{2t}(G))$ and $\chi(M^{2t}(G))$ will also be very close. Before proving the theorem, we need to name the vertices of $M^2(G)$. The naming system shown in Figure 1 is provided in [3], where $\{x^0: x \in V(G)\}$ induces the original graph G. For any vertex $x \in V(G)$, x^1 is the twin of x in $M(G)$, u^1 is the root of $M(G)$. Moreover, x^2 is the twin of x^0 in $M^2(G)$, x^3 is the twin of x^1 in $M^2(G)$, and u^2 is the root of $M^2(G)$.

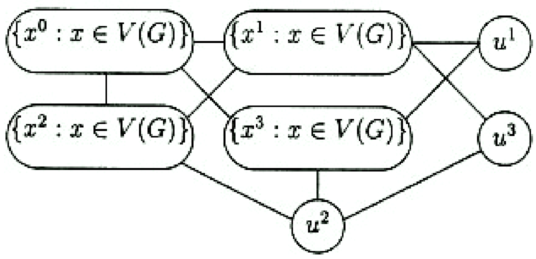

Figure 1. $M^2(G)$

Theorem 3.2 Let t be some positive integer with $t \geq 2$. Then $\chi_c(M^2(K_3^{3t+1})) \leq (\chi(M^2(K_3^{3t+1})) - 1) + 1/3$.

Proof It is known that the graph K_3^{3t+1} has $\chi_c(K_3^{3t+1}) = t + 1/3 = (\chi(K_3^{3t+1}) - 1) + 1/3$. Let c: $V(K_3^{3t+1}) \rightarrow \{0, 1, 2, \ldots, 3t\}$ be a $(3t+1, 3)$-coloring of G. Define c': $V(M^2(K_3^{3t+1})) \rightarrow \{0, 1, 2, \ldots, 3t+6\}$ as

$$c'(x^i) = \begin{cases} 3t+5 & \text{if } i=0 \text{ and } x=3t-1 \\ x+3t+1 & \text{if } i=2 \text{ and } x \leq 1 \\ 3t+1 & \text{if } i=3 \text{ and } x \leq 1 \\ x & \text{otherwise} \end{cases}$$

and $c'(u^1) = 3t+4$, $c'(u^3) = 3t+3$, $c'(u^2) = 3t+6$. It is straightforward to verify that $3 \leq |c'(a) - c'(b)| \leq 3t + 4$ for each $ab \in E(M^2(K_3^{3t+1}))$. Hence c' is a $(3t+7, 3)$-coloring of $M^2(K_3^{3t+1})$. ■

Corollary 3.3 If G is a graph for which $\chi_c(G) \leq (\chi(G)-1)+1/3$, then $\chi_c(M^2(G)) \leq (\chi(M^2(G)) - 1)+1/3$. Moreover, $\chi_c(M^{2t}(G)) \leq (\chi(M^{2t}(G)) - 1) + 1/3$ for any integer $t \geq 2$.

Proof Since G has a homomorphism to K_3^{3t+1}, $M^2(G)$ also has a homomorphism to $M^2(G)$. The conclusion follows from Theorem 3.2. ■

Let t be some positive integer. Now we consider the graphs $M^2(K_4^{4t+3})$. If $t = 2$, the graph $M^2(K_4^{11})$ has circular chromatic number $\chi_c(M^2(K_4^{11})) \leq \chi(M^{2t}(K_4^{11})) - 1/2$ by the result $\chi_c(M^{n-1}(K_n)) \leq \chi(M^{n-1}(K_n)) - 1/2$ (provided in [3]). If $t \geq 3$, it seems impossible to find a $(4t+11, 4)$-coloring of the graph $M^2(K_4^{4t+3})$. Furthermore, when $t = 3$, it seems impossible to find an acyclic 6-coloring (cf. [7]) for $M^2(K_4^{15})$. So we make the following conjecture:

Conjecture If k, d are relatively prime integers, $d \geq 4$, $k > 3d$ and $\chi_c(K_d^k) > \chi(K_d^k) - 1/3$, then $\chi_c(M^2(K_d^k)) = \chi(M^2(K_d^k))$.

4. Conclusions

In this paper, we studied some properties of the circular chromatic numbers of the graphs $M^t(G)$. The upper bound of d when $\chi_c(M^t(G)) = k/d$ is improved, especially for $G=K_n$ or K_d^k. We also proved that if $\chi_c(G) \leq (\chi(G) - 1) + 1/3$, then $\chi_c(M^{2t}(G)) \leq (\chi(M^{2t}(G)) - 1) + 1/3$ for every positive integer t. Although it is known that if $\chi_c(G) \leq \chi(G) - 1/3$, then $\chi_c(M^2(G)) \leq \chi(M^2(G)) - 1/3$. It seems impossible to improve the result from 1/3 to 1/4. Therefore we conjecture that if $\chi_c(K_d^k)$ and $\chi(K_d^k)$ are close enough, then $\chi_c(M^2(K_d^k)) = \chi(M^2(K_d^k))$.

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